

Problems

- Prove that average momentum of a particle in a box is zero.

$$\langle p \rangle = 0$$

Solution:

we know $\psi(x) = \sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l}$

probability distribution, $\psi^*(x) \psi(x) dx = \frac{2}{l} \sin^2 \frac{n\pi x}{l} dx$

The average value of x , or the mean position of the particle is given by,

$$\begin{aligned}
 \langle x \rangle &= \frac{2}{l} \int_0^l x \sin^2 \frac{n\pi x}{l} dx \\
 &= \frac{2}{l} \times \frac{l^2}{4} \\
 &= \frac{l}{2}
 \end{aligned}$$

Similarly, $\langle x^2 \rangle = \frac{2}{l} \int_0^l x^2 \sin^2 \frac{n\pi x}{l} dx$

$$= \left(\frac{l}{2\pi n} \right)^2 \left(\frac{4\pi^2 n^2}{3} - 2 \right)$$

$$= \frac{l^2}{3} - \frac{l^2}{2n^2\pi^2}$$

The variance of x ,

$$\begin{aligned}\sigma_x^2 &= \langle x^2 \rangle - \langle x \rangle^2 \\ &= \left(\frac{l}{2\pi n} \right)^2 \left(\frac{\pi^2 n^2}{3} - 2 \right)\end{aligned}$$

So, Std. Dev.

$$\sigma_x = \frac{l}{2\pi n} \left(\frac{\pi^2 n^2}{3} - 2 \right)^{1/2}$$

$$H\psi = E\psi$$

multiply with ψ^* and integrate

$$\int \psi^* H\psi dx = \int \psi^* E\psi dx$$

$$= E \int \psi^* \psi dx$$

$$= E$$

$$\left[\int \psi^* \psi dx = 1 \right]$$

We know:

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

$$P_x = -i\hbar \frac{d}{dx}$$

$$\langle S \rangle = \int \psi^*(x) \hat{S} \psi(x) dx$$

$$\begin{aligned} \text{So, } \langle p \rangle &= \int_0^L \left[\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \right] \left(-i\hbar \frac{d}{dx} \right) \left[\sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \right] dx \\ &= -i\hbar \frac{2\pi n}{L^2} \underbrace{\int_0^L \sin \frac{n\pi x}{L} \cos \frac{n\pi x}{L} dx}_0 \end{aligned}$$

$$\text{So, } \langle p \rangle = 0$$

variance in momentum.

$$\sigma_p^2 = \langle p^2 \rangle - \langle p \rangle^2$$

$$\begin{aligned}\langle p^2 \rangle &= \int \psi^*(x) \hat{p}_x^2 \psi(x) dx \\&= \int_0^l \left[\sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l} \right] \left(-\hbar^2 \frac{d^2}{dx^2} \right) \left[\sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l} \right] dx \\&= \frac{2n^2 \pi^2 \hbar^2}{l^3} \int_0^l \sin \frac{n\pi x}{l} \sin \frac{n\pi x}{l} \\&= \frac{2n^2 \pi^2 \hbar^2}{l^3} \times \frac{l}{2} = \frac{n^2 \pi^2 \hbar^2}{l^2}\end{aligned}$$

Again,

$$\begin{aligned}
 \langle E \rangle &= \left\langle \frac{p^2}{2m} \right\rangle \\
 &= \frac{\langle p^2 \rangle}{2m} \\
 &= \frac{n^2 \pi^2 \hbar^2}{2m l^2}
 \end{aligned}$$

$$\sigma_p^2 = \frac{n^2 \pi^2 \hbar^2}{l^2}$$

or, $\sigma_p = \frac{n\pi\hbar}{l}$

or, $\sigma_p \propto \frac{1}{l}$
 $\sigma_x \propto l$

Now, $\sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{\left(\frac{\pi^2 n^2}{3} - 2 \right)}$

So, $\sigma_x \sigma_p > \frac{\hbar}{2}$

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